

Matthew Reyna

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Active U.S. Secret Security Clearance

EDUCATION

Texas Tech University, Lubbock, Texas
Bachelor of Science in Computer Engineering

December 2026

GPA: 3.5

RELEVANT COURSE WORK

- Control System Analysis & Design
- Electronics I & II
- Signals & Systems
- Microprocessor Architecture

PROJECT EXPERIENCE

Digital Audio Post Processor, Lubbock, Texas

Digital Communications Project Lab

January 2026 – May 2026

- Designed and developed a digital audio processing system with real-time DSP features.
- Implemented digital filters, 3-band equalizer, FFT and live waveform visualization for audio analysis and processing.
- Collaborated with a three-person engineering team and demonstrated our system on hardware and software.

Smart Home Energy Monitoring & Control System, Lubbock, Texas

Embedded Systems Capstone Project

November 2025 – December 2025

- Validated a STM32/ESP32 platform that measured voltage, current, and power from PZEM sensors.
- Troubleshoot ESP32 Wi-Fi and UART telemetry, verifying end-to-end data transfer.
- Provided 24-hour fully functional prototype that monitors common household devices.

Basketball Shot Accuracy System, Lubbock, Texas

Microcontrollers Project Lab

August 2025 – November 2025

- Developed embedded platform with basketball shot detection and wireless event detection.
- Built hardware Hall sensor circuit, 3D printed housing and software data processing.
- Demonstrated reliable real-time scoring during live testing.

Autonomous Line-Following Rover, Lubbock, Texas

FPGA Project Lab

January 2025 – May 2025

- Implemented Verilog state machines on Basys3 for autonomous robot warehouse navigation.
- Developed different modules for motor control, line following, and signal processing.
- Presented a functional rover that accomplished all objectives within given engineering constraints.

WORK EXPERIENCE

Texas Tech University ECE Department, Lubbock, Texas

Electronics I Tutor

August 2025 – December 2025

- Guided students through circuit analysis, biasing, and measurement concepts, reinforcing oscilloscope and LTspice validation methods.
- Collaborated with students to solve homework problems and LTspice design projects.

INVOLVEMENT

LYFE (Launch Your Future in Engineering)

Engineering Mentor

August 2024 - Present

- Provided guidance and support to first year engineering students.
- Influenced future generations of students to pursue a successful career in engineering.

TECHNICAL SKILLS

- Security Clearance: Active Secret (Granted: August 2026)
- Embedded: C/C++, Python, Verilog, Vivado, Linux, Virtual Machines, MATLAB, RTOS concepts.
- Tools: Xilinx, LTspice, oscilloscope, DMM, DC power supply, soldering iron, FPGA, OnShape.
- Interfaces: UART, I2C, BLE